

Total no. of Pages:

Roll no.

Degree: B. Tech (CSIOT & CSDA) Semester: II

END-SEMESTER EXAMINATION, APRIL-MAY 2025

Course Code: CBEP03/CIEPC03 (Reappear)

Course Title: DIGITAL LOGIC DESIGN

Max. Marks: 40

Duration: 03 Hours

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	State D-morgan's laws and verify using truth table.	4	1
1b	Given numbers: $x = 164$ and $y = 75$; (1) find binary and hexadecimal value of x and y (2) evaluate $(x - y)$ using 2's complement.	2 + 2	1
1c	Given numbers: $p = 265$ and $q = 973$; convert them to BCD and find their BCD sum.	4	1
Q 2	Attempt any 2 parts of the following		
2a	Minimize the Boolean function using K-map and implement using only NAND gates:- $F(A, B, C, D) = \sum \min (1, 3, 6, 11, 12, 14) + d (0, 4, 9, 15)$.	4	2
2b	Design a BCD adder using two 4-bit parallel adders.	4	2
2c	Using an 8:1 MUX, implement the function: $F(P, Q, R, S) = \sum \min (0, 1, 4, 7, 8, 11, 12, 14)$	4	2
Q 3	Attempt any 2 parts of the following		
3a	Draw and explain 4-bit PISO shift register, using JK Flip-flops	4	3
3b	Design a 3-bit synchronous down counter using D Flip-flops	4	3
3c	Design a logic circuit to detect the presence of the 4-bit sequence: 0 1 1 0	4	3
Q 4	Attempt any 2 parts of the following		
4a	Write a VHDL module code for: (a) JK-F/F and (b) D-F/F	2 + 2	4
4b	Write a VHDL module code 4-bit binary counter	4	4
4c	Write a VHDL module code for 8:1 Multiplexer	4	4
Q 5	Attempt any 2 parts of the following		
5a	Explain the following parameters of logic families: (i) Propagation delay, (ii) Noise margin, (iii) fan-in and fan-out	4	5
5b	Draw and explain CMOS transistor level schematic of 3-inputs NOR gate	4	5
5c	Write a short note on PLA	4	5